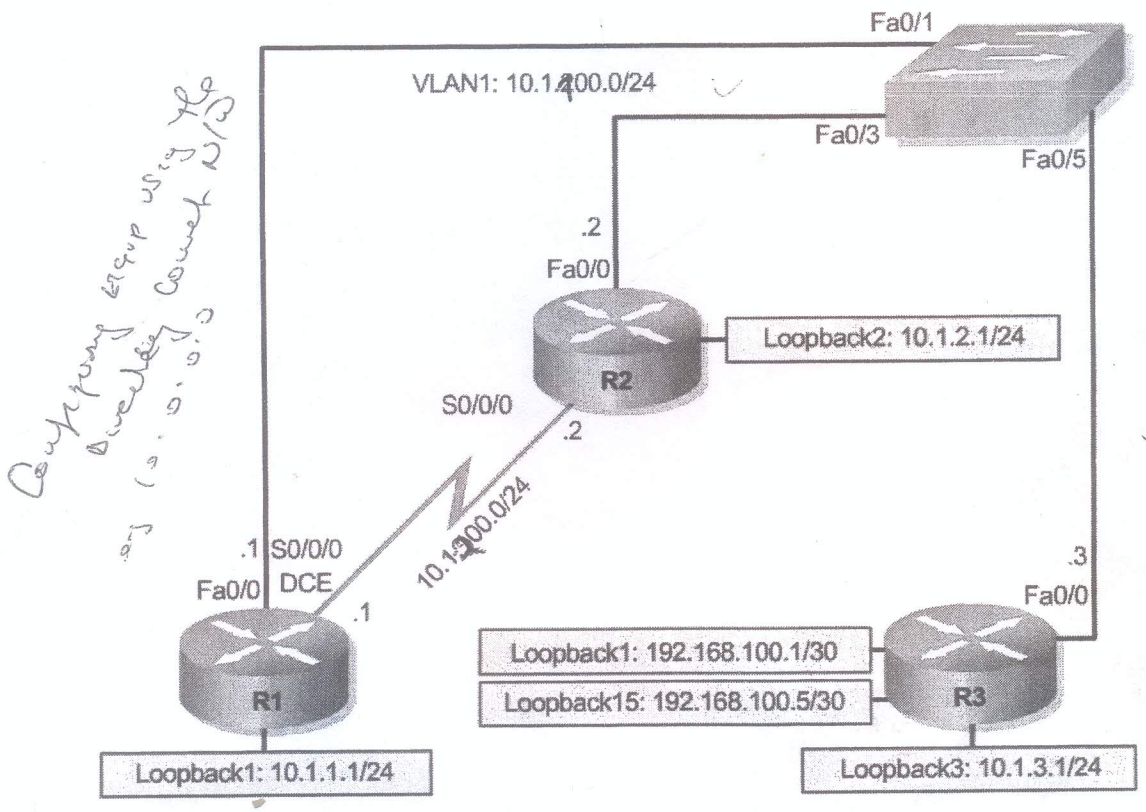
Lab 2-1 EIGRP Configuration, Bandwidth, and Adjacencies

Learning Objectives

- Configure EIGRP on an interface
- Configure the bandwidth command to limit EIGRP bandwidth
- Verify EIGRP adjacencies
- Verify EIGRP routing information exchange
- Utilize debugging commands for troubleshooting EIGRP
- Challenge: Test convergence for EIGRP when a topology change occurs

Topology Diagram



Scenario

You are responsible for configuring the new network to connect your company's Engineering, Marketing, and Accounting departments, represented by the loopback interfaces on each of the three routers. The physical devices have just

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been installed and are connected by Fast Ethernet and serial cables. Your task is to configure EIGRP to enable full connectivity between all departments.

Step 1: Addressing

Using the addressing scheme in the diagram, apply IP addresses to the Fast Ethernet interfaces on R1, R2, and R3. Then create Loopback1 on R1, Loopback2 on R2, and Loopback3 on R3 and address them according to the diagram.

```
R1#configure terminal
R1(config)# interface Loopback1
R1(config-if)# description Engineering Department
R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# exit
R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 10.1.100.1 255.255.255.0

R2#configure terminal
R2(config)# interface Loopback2
R2(config-if)# description Marketing Department
R2(config-if)# ip address 10.1.2.1 255.255.255.0
R2(config-if)# exit
R2(config)# interface FastEthernet0/0
R2(config-if)# ip address 10.1.100.2 255.255.255.0

R3#configure terminal
R3(config)# interface Loopback3
R3(config-if)# description Accounting Department
R3(config-if)# ip address 10.1.3.1 255.255.255.0
R3(config-if)# exit
R3(config)# interface FastEthernet0/0
R3(config-if)# ip address 10.1.100.3 255.255.255.0
```

Leave the switch in its default (blank) configuration. By default, all switch ports are in VLAN1 and are not administratively down.

For now, also leave the serial interfaces in their default configuration. You will configure the serial link between R1 and R2 in Step 4.

Verify that the line protocol of each interface is up and that you can successfully ping across each link. You should see similar output on each router:

```
R1#sh ip interface brief
Interface                  IP-Address      OK? Method Status
Protocol
FastEthernet0/0            10.1.100.1      YES manual up                    up
FastEthernet0/1            unassigned      YES unset  administratively down
down
Serial0/0/0                unassigned      YES manual up                    up
Serial0/0/1                unassigned      YES unset  administratively down
down
Loopback1                  10.1.1.1        YES manual up                    up
```

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Step 2: Configuring EIGRP Across VLAN1

After you have implemented your addressing scheme, create an EIGRP autonomous system (AS) on R1 using the following commands in global configuration mode:

```
R1(config)# router eigrp 1
R1(config-router)# network 10.0.0.0
```

Using network statements with major networks causes EIGRP to begin sending EIGRP Hello packets out all interfaces in that network (that is, subnets of the major network 10.0.0.0/8). In this case, EIGRP should start sending Hello packets out of its Fast Ethernet and loopback interfaces. To check if this is occurring, use the **debug eigrp packets** in privileged-exec mode.

```
R1#debug eigrp packets
*Sep 25 21:27:09.547: EIGRP: Sending HELLO on Loopback1
*Sep 25 21:27:09.547:   AS 1, Flags 0x0, Seq 0/0 idbQ 0/0 iidbQ un/rely 0/0
*Sep 25 21:27:09.547: EIGRP: Received HELLO on Loopback1 nbr 10.1.1.1
*Sep 25 21:27:09.547:   AS 1, Flags 0x0, Seq 0/0 idbQ 0/0
*Sep 25 21:27:09.547: EIGRP: Packet from ourselves ignored
*Sep 25 21:27:10.203: EIGRP: Sending HELLO on FastEthernet0/0
*Sep 25 21:27:10.203:   AS 1, Flags 0x0, Seq 0/0 idbQ 0/0 iidbQ un/rely 0/0
R1#undebug all
```

These Hello packets are unanswered by the other routers, because EIGRP is not yet running on R2 or R3. R1 ignores the Hello packets from itself on Loopback1. Use the **undebug all** command to stop the debug output.

Which interfaces are involved in EIGRP's routing process on this router? Use **show ip eigrp interfaces** to show which interfaces are participating in EIGRP. You should see output similar to the following: lo1 loopback1 & Fa0/0 are participating in EIGRP routing process on R1

```
R1#show ip eigrp interfaces
IP-EIGRP interfaces for process 1

                      Xmit Queue   Mean   Pacing Time   Multicast    Pending
Interface        Peers  Un/Reliable  SRTT   Un/Reliable   Flow Timer   Routes
Fa0/0              0        0/0        0        0/1           0           0
Lo1                0        0/0        0        0/1           0           0
```

You are interested in seeing the adjacency initiate on R1 and R2, so you issue **debug eigrp packets** on R1 and R2 to monitor the adjacency taking place in real time while you configure R2.

Now, in global configuration mode on R2, issue the same set of commands you issued on R1 to create EIGRP AS 1 and advertise the 10.0.0.0/8 network. You should see debug output similar to the following:

```
R2#debug eigrp packets
EIGRP Packets debugging is on
    (UPDATE, REQUEST, QUERY, REPLY, HELLO, IPXSAP, PROBE, ACK, STUB, SIAQUERY,
SIAREPLY)
```

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```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router eigrp 1
R2(config-router)#network 10.0.0.0
R2(config-router)#
*Sep 25 20:32:28.427: EIGRP: Sending HELLO on FastEthernet0/0
*Sep 25 20:32:28.427:   AS 1, Flags 0x0, Seq 0/0 idbQ 0/0 iidbQ un/rely 0/0
*Sep 25 20:32:28.431: EIGRP: Received HELLO on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.431:   AS 1, Flags 0x0, Seq 0/0 idbQ 0/0
*Sep 25 20:32:28.431: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor 10.1.100.1
(FastEthernet0/0) is up: new adjacency
*Sep 25 20:32:28.431: EIGRP: Enqueueing UPDATE on FastEthernet0/0 nbr
10.1.100.1 iidbQ un/rely 0/1 peerQ un/rely 0/0
*Sep 25 20:32:28.435: EIGRP: Received UPDATE on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.435:   AS 1, Flags 0x1, Seq 1/0 idbQ 0/0 iidbQ un/rely 0/1
peerQ un/rely 0/0
*Sep 25 20:32:28.435: EIGRP: Requeued unicast on FastEthernet0/0
*Sep 25 20:32:28.435: EIGRP: Sending HELLO on FastEthernet0/0
*Sep 25 20:32:28.435:   AS 1, Flags 0x0, Seq 0/0 idbQ 0/0 iidbQ un/rely 0/0
*Sep 25 20:32:28.439: EIGRP: Sending UPDATE on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.439:   AS 1, Flags 0x1, Seq 1/1 idbQ 0/0 iidbQ un/rely 0/0
peerQ un/rely 0/1
*Sep 25 20:32:28.443: EIGRP: Received UPDATE on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.443:   AS 1, Flags 0x8, Seq 2/0 idbQ 0/0 iidbQ un/rely 0/0
peerQ un/rely 0/1
*Sep 25 20:32:28.447: EIGRP: Received ACK on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.447:   AS 1, Flags 0x0, Seq 0/1 idbQ 0/0 iidbQ un/rely 0/0
un/rely 0/1
*Sep 25 20:32:28.447: EIGRP: Enqueueing UPDATE on FastEthernet0/0 nbr
10.1.100.1 iidbQ un/rely 0/1 peerQ un/rely 0/0 serno 1-2
*Sep 25 20:32:28.451: EIGRP: Requeued unicast on FastEthernet0/0
*Sep 25 20:32:28.455: EIGRP: Sending UPDATE on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.455:   AS 1, Flags 0x8, Seq 2/2 idbQ 0/0 iidbQ un/rely 0/0
peerQ un/rely 0/1 serno 1-2
*Sep 25 20:32:28.455: EIGRP: Enqueueing UPDATE on FastEthernet0/0 iidbQ
un/rely 0/1 serno 3-3
*Sep 25 20:32:28.455: EIGRP: Received UPDATE on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.455:   AS 1, Flags 0x8, Seq 3/1 idbQ 0/0 iidbQ un/rely 0/1
peerQ un/rely 0/1
*Sep 25 20:32:28.455: EIGRP: Enqueueing ACK on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.455:   Ack seq 3 iidbQ un/rely 0/1 peerQ un/rely 1/1
*Sep 25 20:32:28.459: EIGRP: Received ACK on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.459:   AS 1, Flags 0x0, Seq 0/2 idbQ 0/0 iidbQ un/rely 0/1
peerQ un/rely 1/1
*Sep 25 20:32:28.467: EIGRP: Forcing multicast xmit on FastEthernet0/0
*Sep 25 20:32:28.467: EIGRP: Sending UPDATE on FastEthernet0/0
*Sep 25 20:32:28.467:   AS 1, Flags 0x0, Seq 3/0 idbQ 0/0 iidbQ un/rely 0/0
serno 3-3
*Sep 25 20:32:28.471: EIGRP: Received ACK on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.471:   AS 1, Flags 0x0, Seq 0/3 idbQ 0/0 iidbQ un/rely 0/0
peerQ un/rely 1/1
*Sep 25 20:32:28.471: EIGRP: FastEthernet0/0 multicast flow blocking cleared
*Sep 25 20:32:28.479: EIGRP: Sending ACK on FastEthernet0/0 nbr 10.1.100.1
*Sep 25 20:32:28.479:   AS 1, Flags 0x0, Seq 0/3 idbQ 0/0 iidbQ un/rely 0/0
peerQ un/rely 1/0
```

The debug output displays the EIGRP Hello, Update, and ACK packets. Because EIGRP uses Reliable Transport Protocol (RTP) for Update packets, you see routers replying to Update packets with the ACK packet. You can turn off debugging with **undebug all**.

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Configure EIGRP on R3 using the same commands.

```
R3(config)# router eigrp 1
R3(config-router)# network 10.0.0.0
```

Step 3: Verifying the EIGRP Configuration

When R3 is configured, issue **show ip eigrp neighbors** on each router. If you have configured each router successfully, there are two adjacencies on each router.

```
R1#show ip eigrp neighbors
IP-EIGRP neighbors for process 1
H   Address                 Interface       Hold Uptime   SRTT   RTO  Q  Seq
                                            (sec)         (ms)       Cnt Num
1   10.1.100.3              Fa0/0             10 00:00:17    1   200  0  7
0   10.1.100.2              Fa0/0             11 00:02:01    5   200  0  6
!
R2#sh ip eigrp neighbors
IP-EIGRP neighbors for process 1
H   Address                 Interface       Hold Uptime   SRTT   RTO  Q  Seq
                                            (sec)         (ms)       Cnt Num
1   10.1.100.3              Fa0/0             13 00:00:56    1   200  0  7
0   10.1.100.1              Fa0/0             12 00:02:40    1   200  0  47
!
R3#sh ip eigrp neighbors
IP-EIGRP neighbors for process 1
H   Address                 Interface       Hold Uptime   SRTT   RTO  Q  Seq
                                            (sec)         (ms)       Cnt Num
1   10.1.100.2              Fa0/0             11 00:01:21  819  4914  0  6
0   10.1.100.1              Fa0/0             11 00:01:21    2   200  0  47
```

Now check whether the EIGRP routes are being exchanged between the routers using **show ip eigrp topology**:

```
R1#show ip eigrp topology
IP-EIGRP Topology Table for AS(1)/ID(10.1.1.1)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 10.1.3.0/24, 1 successors, FD is 156160
        via 10.1.100.3 (156160/128256), FastEthernet0/0
P 10.1.2.0/24, 1 successors, FD is 156160
        via 10.1.100.2 (156160/128256), FastEthernet0/0
P 10.1.1.0/24, 1 successors, FD is 128256
        via Connected, Loopback1
P 10.1.100.0/24, 1 successors, FD is 28160
        via Connected, FastEthernet0/0
```

You should see all the networks currently advertised by EIGRP on every router. We will explore the output of this command in the next lab. For now, verify that each of the loopback networks exist in the EIGRP topology table. Because EIGRP is the only routing protocol running and currently has routes to these networks, issuing **show ip route eigrp** displays the best route to the destination network.

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```
R1#show ip route eigrp
     10.0.0.0/24 is subnetted, 4 subnets
D       10.1.3.0 [90/156160] via 10.1.100.3, 00:00:53, FastEthernet0/0
D       10.1.2.0 [90/156160] via 10.1.100.2, 00:00:53, FastEthernet0/0
```

To check whether you have full connectivity, ping the remote loopbacks from each router. If you have successfully pinged all the remote loopbacks, congratulations! You have configured EIGRP to route between these three remote networks.

Step 4: Configuring EIGRP on the Serial Interfaces

Your serial interfaces are still be in their default configuration. Address the interface according to the diagram, and set the clock rate to 64 kbps.

```
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.1.200.1 255.255.255.0
R1(config-if)# clock rate 64000
R1(config-if)# no shut
!
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.1.200.2 255.255.255.0
R2(config-if)# no shut
```

Notice that even though you have clocked the interface at 64 kbps, issuing **show interface serial 0/0/0** reveals that the interface still is a full T1 bandwidth of 1544 kbps.

```
R1#show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is up
  Hardware is GT96K Serial
  Internet address is 10.1.200.1/24
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
...
```

By default, EIGRP uses up to 50 percent of the bandwidth that your interface reports to the Cisco IOS software. Suppose there was a significant routing instability in some other part of our EIGRP AS. If EIGRP were to use 50 percent of 1544 kbps for its own routing information traffic, EIGRP traffic would fully saturate our measly 64 kbps serial link!

Also, recall that EIGRP makes bandwidth computations using a composite metric in which one of the variables is the bandwidth of the interface. For EIGRP to make an accurate computation, it needs correct information about the bandwidth of your serial link. Therefore, you need to manually configure the bandwidth variable to 64 kbps. Apply the **bandwidth 64** command to the R1 and R2 serial interfaces as follows:

```
R1:
!
interface Serial0/0/0
 bandwidth 64
!
```

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```
R2:
!
interface Serial0/0/0
 bandwidth 64
!
```

Verify that your bandwidth configuration is reflected in the **show interface serial 0/0/0** output:

```
R1#show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is up
  Hardware is GT96K Serial
  Internet address is 10.1.200.1/24
  MTU 1500 bytes, BW 64 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
...
R2#show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is up
  Hardware is GT96K Serial
  Internet address is 10.1.200.2/24
  MTU 1500 bytes, BW 64 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
...
```

Now, issue the **show ip eigrp neighbors** command, which displays the following neighbor relationship between R1 and R2:

```
R1#show ip eigrp neighbors
IP-EIGRP neighbors for process 1
H   Address                 Interface       Hold Uptime   SRTT   RTO  Q  Seq
                                            (sec)         (ms)       Cnt Num
2   10.1.200.2              Se0/0/0           10 00:03:03   24   200  0  53
1   10.1.100.2              Fa0/0             14 09:22:42  269  1614  0  54
0   10.1.100.3              Fa0/0             11 09:22:42  212  1272  0  59
```

Step 5: Configuring Network Statement Wildcard Masks

On R3, create Loopback11 with IP address 192.168.100.1/30, and Loopback15 with IP address 192.168.100.5/30.

```
R3(config)# interface Loopback11
R3(config-if)# ip address 192.168.100.1 255.255.255.252
R3(config-if)# exit
R3(config)# interface Loopback15
R3(config-if)# ip address 192.168.100.5 255.255.255.252
R3(config-if)# exit
```

How can you add the 192.168.100.0/30 network to EIGRP without involving the 192.168.100.4/30 network as well?

In Step 2, we looked at how network statements select networks for routing using major network boundaries. EIGRP also provides a way to select networks using wildcard masks. In a wildcard mask, bits that may vary are denoted by 1s in the binary bit values. If we wanted to route both Loopback11 and Loopback15 with EIGRP, we could use a wildcard mask that includes both of their network addresses, such as **network 192.168.100.0 0.0.0.7** or **network**

> Use a mask in the EIGRP N/W statement to involve the 192.168.100.0/30 subnet & not 192.168.100.4/30 subnet.

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this command is network 192.168.100.0 0.0.0.3
this command will allow any values for the last 2 bits

192.168.100.0 0.0.0.255. However, in this scenario, we only want to select Loopback11's IP network.

On R3, issue the following commands:

```
R3(config)# router eigrp 1
R3(config-router)# network 192.168.100.0 0.0.0.3
```

Did this solution work? Check it with the **show ip eigrp interfaces** command. Notice that Loopback11 is involved in EIGRP, and Loopback15 is not.

```
R3#show ip eigrp interfaces
IP-EIGRP interfaces for process 1

                      Xmit Queue   Mean   Pacing Time   Multicast    Pending
Interface        Peers Un/Reliable  SRTT   Un/Reliable   Flow Timer   Routes
Fa0/0              2        0/0       5        0/1           50           0
Lo3                0        0/0       0        0/1            0           0
Lo11               0        0/0       0        0/1            0           0
```

Which of these two IP networks can you see in the routing table on R1 after EIGRP converges with the new network? Look at the output of **show ip route eigrp** on R1.

```
R1#show ip route eigrp
     10.0.0.0/24 is subnetted, 5 subnets
D       10.1.3.0 [90/156160] via 10.1.100.3, 00:05:59, FastEthernet0/0
D       10.1.2.0 [90/156160] via 10.1.100.2, 00:12:16, FastEthernet0/0
D    192.168.100.0/24 [90/156160] via 10.1.100.3, 00:03:05, FastEthernet0/0
```

Notice that the subnet mask for the 192.168.100.0 network advertised by R3 is 24 bits. This will be examined in much further depth in the next lab. Do you remember the command to allow R3 to advertise the proper subnet mask to its adjacent routers? If so, record it below:

network 192.168.100.0 0.0.0.3

Challenge: Topology Change

You have been reading up about the advantages of different routing protocols in your spare time. You noticed statements claiming that EIGRP converges significantly faster than other routing protocols in a topology where there are multiple paths to the destination network. You are interested in testing this before you bring the network that you are designing online.

Verify that all the neighbor relationships are active and that the routing tables of each router have the original three loopback interfaces of the other routers as described in the initial diagram. Make sure you issue the **debug ip eigrp 1** command on all routers. You are not going to want to miss this!

You have observed the following output:

```
R2#show ip route eigrp
     10.0.0.0/24 is subnetted, 5 subnets
```

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```
D       10.1.3.0 [90/156160] via 10.1.100.3, 00:05:22, FastEthernet0/0
D       10.1.1.0 [90/156160] via 10.1.100.1, 00:05:22, FastEthernet0/0
!
R3#sh ip route eigrp
     10.0.0.0/24 is subnetted, 5 subnets
D       10.1.2.0 [90/156160] via 10.1.100.2, 09:25:37, FastEthernet0/0
D       10.1.1.0 [90/156160] via 10.1.100.1, 09:25:37, FastEthernet0/0
D       10.1.200.0 [90/40514560] via 10.1.100.2, 00:03:01, FastEthernet0/0
                   [90/40514560] via 10.1.100.1, 00:03:01, FastEthernet0/0
!
R3#traceroute 10.1.1.1

Type escape sequence to abort.
Tracing the route to 10.1.1.1

  1 10.1.100.1 4 msec *  0 msec
R3#
```

R3 is using R1 as the next hop to get to destination network 10.1.1.0/24 per R3's routing table. However, R3 could potentially get to R1 through R2 via the serial link if the Fast Ethernet port on R1 was shut down.

From R3, issue a **ping** with a high repeat count to destination address 10.1.1.1:

```
ping 10.1.1.1 repeat 100000
```

You should see multiple exclamation points flooding the console output from R3. On R1, shut down the FastEthernet0/0 interface:

```
R3(config)# interface FastEthernet0/0
R3(config-if# shutdown
```

From R3's perspective, how many packets were dropped? Which of the EIGRP timers causes this delay in the route recalculation?

Use the **traceroute** tool to find the new route from R3 to R1:

```
R3#traceroute 10.1.1.1

Type escape sequence to abort.
Tracing the route to 10.1.1.1

  1 10.1.100.2 0 msec 4 msec 0 msec
  2 10.1.200.1 12 msec *  12 msec
```

Start your repeated ping again from R3 and administratively open the FastEthernet0/0 interface on R1 again.

How many packets were dropped when the FastEthernet0/0 interface went up?
Approx 13 packets should be dropped.
If you were using RIPv2 as your routing protocol instead of EIGRP, would fewer packets or more packets be dropped? (BACK PAGE)

When R1's Fa goes down it sends update to R2 which forwards the update to R3. the routers go into active state to search for a new route (if necessary) and recalculate.

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Approx 7 packet should be dropped.

Although u may reason that RIPv2 would have converged faster than a EIGRP, the true reason is that EIGRP converges faster b/c the DUAL algorithm allows diffusing updates to trigger the route recalculation early on. Thus, RIPv2 would send more packets during the convergence than EIGRP.

END OF LAB CONFIGS:

```
R1#sh run
Building configuration...
!
hostname R1
!
interface Loopback1
 description Engineering Department
 ip address 10.1.1.1 255.255.255.0
!
interface FastEthernet0/0
 ip address 10.1.100.1 255.255.255.0
 no shutdown
!
!
interface Serial0/0/0
 bandwidth 64
 ip address 10.1.200.1 255.255.255.0
 clockrate 64000
 no shutdown
!
router eigrp 1
 network 10.0.0.0
 auto-summary
!
end


R2#sh run
Building configuration...
!
hostname R2
!
interface Loopback2
 description Marketing Department
 ip address 10.1.2.1 255.255.255.0
!
interface FastEthernet0/0
 ip address 10.1.100.2 255.255.255.0
 no shutdown
!
interface Serial0/0/0
 bandwidth 64
 ip address 10.1.200.2 255.255.255.0
 no shutdown
!
router eigrp 1
 network 10.0.0.0
 auto-summary
!
end


R3#sh run
Building configuration...
```

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```
!
hostname R3
!
interface Loopback3
 description Accounting Department
 ip address 10.1.3.1 255.255.255.0
!
interface Loopback11
 ip address 192.168.100.1 255.255.255.252
!
interface Loopback15
 ip address 192.168.100.5 255.255.255.252
!
interface FastEthernet0/0
 ip address 10.1.100.3 255.255.255.0
 no shutdown
!
!
router eigrp 1
 network 10.0.0.0
 network 192.168.100.0 0.0.0.3
 auto-summary
!
end
```

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